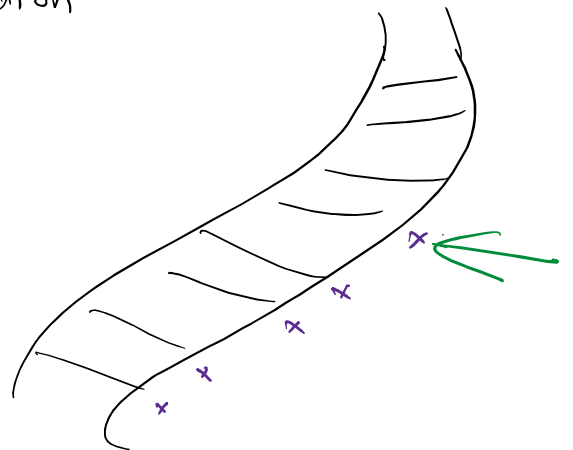


Mathematized Induction

Infinite ladder

Given

- Base Case (i) we can reach rung 1
- Induction step (ii) If we can reach rung i , then we can reach rung $i+1$, for all $i \geq 1$



Continuum hypothesis

Then, we can reach all rungs.

In general form,

Consider a propositional function $P(n)$.

To prove $P(n)$ is TRUE for all +ve integers,

Induction Base Case: Show $P(1)$ is TRUE. ✓

Induction step: $P(k) \rightarrow P(k+1), \forall k \geq 1$

Strong Induction: $(P(1) \wedge P(2) \wedge \dots \wedge P(k)) \rightarrow P(k+1)$

$$\left[(P(1)) \wedge \left(\forall k \in \mathbb{Z}^+, (P(k) \rightarrow P(k+1)) \right) \right] \rightarrow \forall n, P(n).$$

Domain: +ve integers

Example

Example

① Show that

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

Proof: Let $P(n)$: sum of first n +ve integer.

Base Case: $P(1)$

$$LHS = 1$$

$$RHS = \frac{1(1+1)}{2} = 1 \quad \checkmark$$

Induction Step: $\forall k \geq 1, P(k) \rightarrow P(k+1)$

Let k be any +ve integer.

Assume $P(k)$ is TRUE.

Now,

$$\begin{aligned} P(k+1) &= \overbrace{1 + 2 + \dots + k} + (k+1) \\ &= \frac{k(k+1)}{2} + k+1 \\ &= \frac{(k+1)(k+2)}{2} \quad \checkmark \end{aligned}$$

Practice question Show that,

$$(1) \quad 1 + 3 + 5 + \dots + (2n-1) = n^2$$

$$(2) \quad 1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$$

$$(3) \quad n < 2^n, \quad \forall \text{ positive integers } n.$$

(4) Let S be a set with n elements.

$$\text{Show } |\mathcal{P}(S)| = 2^n$$

$\rightarrow$ Powerset (S).

What is wrong with this proof?

Claim: All dogs are of same color.

Let n denote no. of dogs in the world,

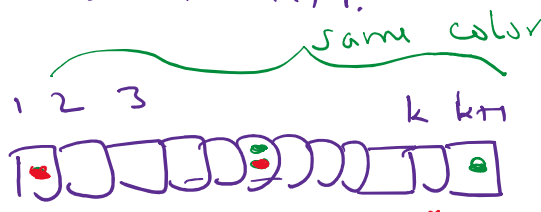
Base Case: $n=1$

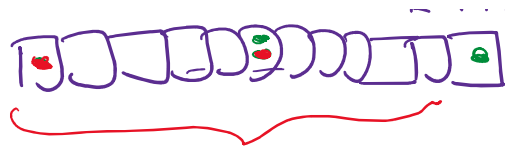
Trivial

Induction Step:

Assume the stmt is TRUE when $n=k$.

Let $n=k+1$.





Same
color



De Morgan Question

Base Case: $n=2$.

$$\overline{A_1 \cap A_2} = \overline{A_1} \cup \overline{A_2} \quad (\text{by De Morgan's law})$$

Ind Step: Assume the stmt is TRUE when $n=k$,
where $k \geq 2$.

Let $n=k+1$.

$$\overline{(A_1 \cap A_2 \cap \dots \cap A_k) \cap A_{k+1}}$$

call it B

$$= \overline{B \cap A_{k+1}}$$

$$= \overline{B} \cup \overline{A_{k+1}} \quad (\text{by De Morgan})$$

$$= \overline{A_1} \cup \overline{A_2} \cup \dots \cup \overline{A_k} \cup \overline{A_{k+1}} \quad (\text{by Ind hyp})$$

Odd pie question . . .

Base Case: $n = 1$. ✓

Ind hyp: Assume it is True when $n = 2k-1$,
where $k \geq 1$.

Ind step: Let $n = 2k+1$.

X

X

X

X

X

$k = 2$

$2k-1 = 3$

X

X

X

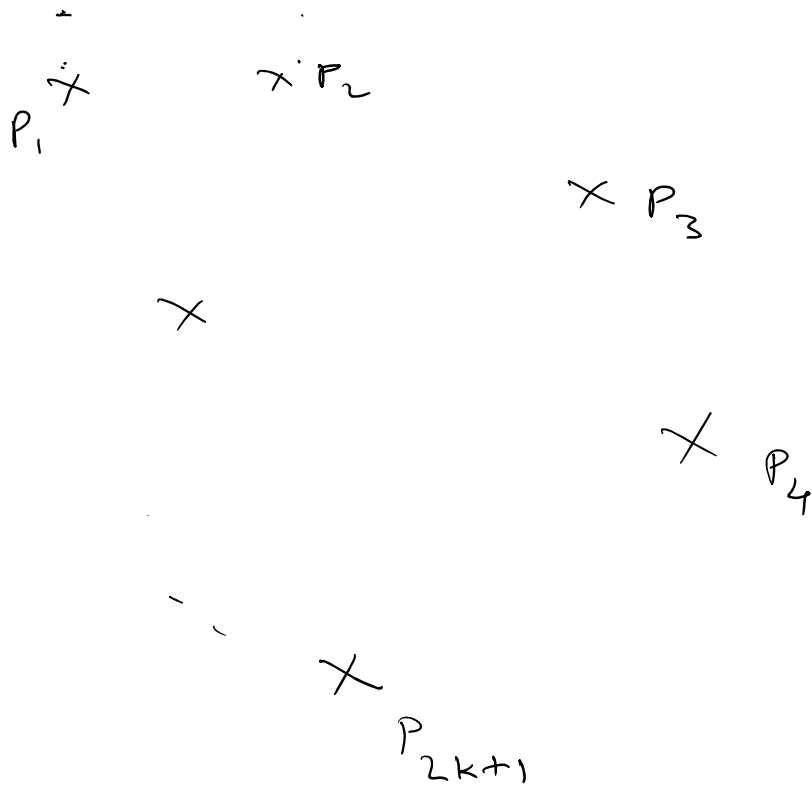
X

$2k+1 = 5$

X

Induction step

Take an instance of size $2k+1$.



Let $\text{dist}(P_i, P_j)$ be the smallest among all pairwise distance. Remove P_i and P_j to obtain an instance of size $2k-1$.

By Induction hypothesis, this smaller instance has a winner, say P_w .

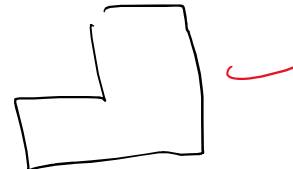
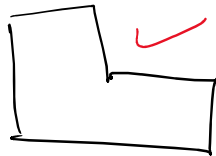
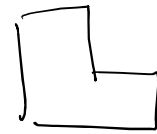
Claim: P_w will be a winner in given full instance of the problem

Proof of claim: When you add back P_i and P_j to the system, neither of them are going to throw at P_w (as they throw at each other).

Triominoes problem

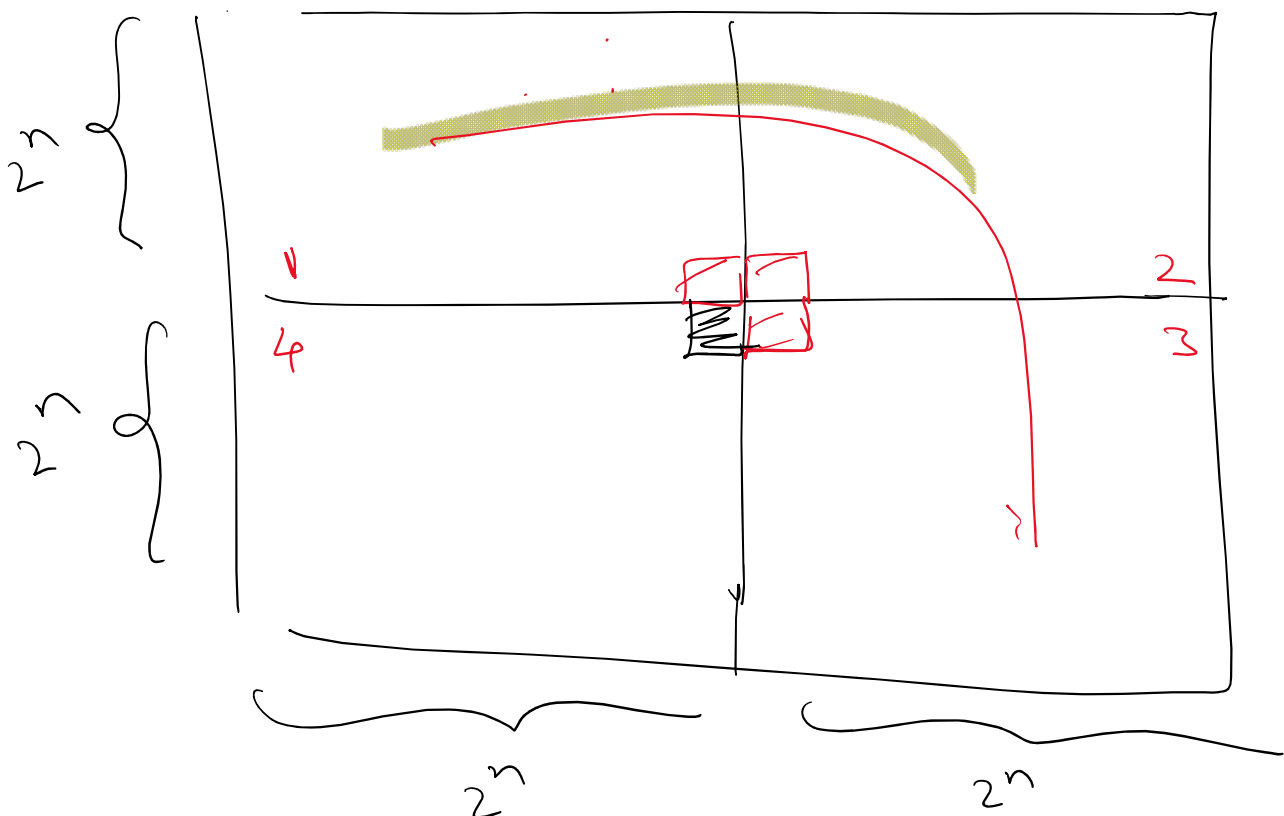
Base Case

$n=1$ $2^1 \times 2^1$



Induction step.

$$2^{n+1} \times 2^{n+1} = (2^n \times 2) \times (2^n \times 2)$$



2''

2''

moved block

Strong Induction

Let n be an integer with $n > 1$.

Q. Let $n > 1$ be an integer. Show that
 n can be written as a product of primes

Proof:

$\hookrightarrow P(n)$

Base Case: $n = 2$.

2 is prime. $\rightarrow P(2)$.

Induction Hypothesis

Assume $P(2), P(3), \dots, P(k)$ is
TRUE, where $k \geq 2$.

Induction step To show $P(k+1)$.

}

Well ordering principle

Any nonempty subset of non-negative integers has a least element.

Q. Prove division algorithm. That is,

Given an integer a and a true integer d , show that there are unique integers q and r with $0 \leq r < d$, such that $a = qd + r$.

Proof:

$$\left\{ \dots, a - \overset{-1}{k_1}d, a - \overset{-2}{k_2}d, a - \overset{-3}{k_3}d, \dots \right\}$$

k_1, k_2, k_3 are integers.

$$\left\{ a - xd : x \in \mathbb{Z}, a - xd \geq 0 \right\}$$

Let $a - qd$ be the smallest element in this set.

$$\text{Let } r = a - qd.$$

Claim: $0 \leq r < d$.

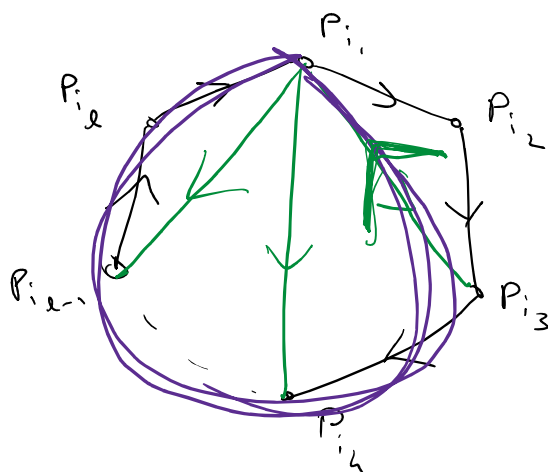
Proof of claim:

from defn of the set constructed,

if $r \geq d$, i.e. if $a - qd \geq d$, then, $a - (q+1)d$ will be an element of the set and it will be smaller than $a - qd$ thus contradicting the claim.

thus contradicting the fact that $a-qd$ was the smallest.

Round robin tournament



→ suppose the smallest cycle had l players with $l > 3$.